



NutriSense

AI-Powered Nutrition, Diet & Clinical Health Companion

Technical User Guide

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Welcome to the NutriSense User Guide

This manual contains detailed step-by-step operating instructions, feature breakdowns, and screenshot guidelines to help you navigate and leverage the app's localized nutritional AI, fasting mode, family profile tracking, clinical workout plans, and predictive coaching.

1. GETTING STARTED & ACCOUNT SETUP

Installation Instructions:

For Android Devices: Locate and download the NutriSense APK (v1.0.0) from the official GitHub repository releases tab. Tap the downloaded file and follow the system prompt to install (allow installation from unknown sources).

Account Sign Up & Sign In:

- Launch the application to open the Welcome Screen.
- Tap "Sign Up" to create a new profile. Enter a valid email address and a strong password. Confirm your password and tap submit.
- Check your inbox for a confirmation email to verify your registration.
- If you already have an account, enter your email and password on the Login Screen and tap "Sign In".
- If you forget your password, tap "Forgot Password?" to receive a secure password reset link via email.

Interactive Onboarding Flow:

On your first login, the app launches an interactive Onboarding dialog.

Step 1: Input your age, gender, height (cm), and weight (kg).

Step 2: Set your physical activity multiplier (Sedentary, Lightly Active, Moderately Active, Very Active).

Step 3: Choose your primary fitness goal (Fat Loss, Muscle Gain, Healthy Lifestyle, or Maintenance).

Step 4: Select any existing medical conditions (Diabetes, Hypertension, PCOS, High Cholesterol, or None).

Step 5: Specify dietary restrictions/allergies (Halal, Vegetarian, Vegan, Nut Allergy, Gluten-Free, Lactose Intolerant, or None).

Tap "Calculate Targets". The system uses Google Gemini to compute your daily calorie goal and water budget, loading them into your profile.

2. EXPLORING THE MAIN DASHBOARD

Dynamic Header:

- At the top of the dashboard, you are greeted by a personalized banner (e.g., "Hello, Affan!") alongside a dynamic calendar date switcher.

Calorie & Nutrition Ring:

- A prominent circular ring displays your calorie progress. It visualizes "Calories Logged" vs. your prescribed "Daily Caloric Budget".
- The center of the ring dynamically displays the remaining calories left for the current day.

Macronutrient Breakdown:

- Directly below the calorie ring, horizontal progress bars track the breakdown of Protein, Carbohydrates, and Fats consumed (in grams) against your daily macro targets.
- Google Health Connect Integration:
- If enabled, the dashboard shows real-time physical telemetry synced from Google Health Connect: Steps taken, Active Calories burned, and Distance traveled.

Today's Meals Timeline:

- Shows a chronological list of meals logged today.
- Each item displays the food description, calculated calories, macro badge, and quick actions to Edit values or Delete the entry.

Hydration Tracker:

- Allows logging water intake with 1-tap logging chips (+250ml cup, +500ml bottle) or entering a custom amount.
- A circular blue ring tracks your hydration progress against your target.

Habit Score Card:

- Displays a rolling habit consistency rating out of 100 with active streak indicators to reward consecutive logging days.

3. LOGGING MEALS WITH AI (ZERO-EFFORT NUTRITION)

Step-by-Step AI Meal Logging:

Step 1: Open the app and tap the "Meals" tab in the bottom navigation bar (or tap the "+ Add Meal" shortcut button on the main dashboard).

Step 2: In the natural language text field, describe what you ate. You can use English, Urdu, or mixed Roman-Urdu phrases (e.g., "1 plate Chicken Biryani and half glass Raita" or "2 Aloo Parathas with 1 cup Chai").

Step 3: Select the Meal Category (Breakfast, Lunch, Dinner, or Snack).

Step 4: (Optional) If you logged a common regional meal, tap any of the quick suggestion chips (e.g., "Egg & Roti", "Daal Chawal") to instantly pre-fill the field.

Step 5: Tap "Log Meal with AI". The system queries Gemini, estimates nutritional metrics based on South Asian cuisine data, and displays the calculated macronutrients in a preview card.

Step 6: Verify the details and tap "Confirm and Log". The meal is saved.

Editing and Correcting AI Values:

- If you need to override the AI's estimation, tap the "Edit Values" button on the preview card.
- Modify the calories, protein, carbs, or fats manually using the numerical input fields, then tap "Save Custom Log".

4. INTERACTING WITH THE AI HEALTH COACH & VOICE ASSISTANT

AI Dietitian Chat:

- Tap the "Coach" tab to start a conversation with your AI Dietitian Coach.
- You can type questions in English or Urdu (e.g., "Explain how to manage hypertension with my diet" or "Suggest a high-protein dinner recipe").
- The AI dietitian processes your query against your profile metrics and medical conditions to return clinically-safe, bulleted markdown guides.

Text-to-Speech Language Playback:

- On any message returned by the AI Coach, tap the green  Speaker icon.
- The voice assistant will read the text aloud. The audio synthesizer dynamically switch pronunciations to read Urdu text in Urdu accent and English text in English accent.

Healthcare Clinic Quick Finder:

- In the AI Coach header, locate the red  Hospital icon.
- Tapping this icon instantly redirects you to the geospatial emergency clinic directory for fast access during health emergencies.

5. RAMADAN FASTING MODE

Activating Ramadan Mode:

- Go to Settings or look at the top of the main dashboard.
- Locate the "Ramadan Fasting Mode" toggle and switch it ON. The dashboard will immediately update with crescent moon motifs, Islamic pattern borders, and specialized widgets.

Sehri & Iftar Timings:

- The dashboard displays a live countdown timer showing the hours, minutes, and seconds remaining until the next meal window (e.g., "Time until Iftar: 3h 15m 12s" or "Time until Sehri: 5h 20m 00s").
- Timings are calculated automatically based on your GPS coordinates.

Ramadan Meal Logging:

- The standard Breakfast, Lunch, and Dinner slots are replaced by Sehri, Iftar, and Post-Taraweeh Snack meal categories.

Ramadan Paced Hydration:

Because hydration is restricted to non-fasting hours, the water tracker presents quick-log buttons aligned with fasting schedules:

- "+500ml Iftar" (Fast Break Hydration)
- "+250ml Taraweeh" (During Prayer Hydration)
- "+500ml Sehri" (Pre-Fast Hydration)

6. MANAGING FAMILY & DEPENDENT PROFILES

Setting Up Dependent Profiles:

- Open the navigation menu and go to Settings -> Family Profiles.
- Tap "+ Add Family Member".
- Enter the family member's Name, select their Relationship (Child, Parent, Spouse, Sibling), and input their age, gender, height, and weight.
- Select their specific medical conditions and dietary restrictions.
- Select a custom avatar color to represent them, then tap "Save Profile".

Logging Meals for Dependents:

- When logging a meal, tap the profile dropdown menu (which defaults to "Myself") and select your family member's name.
- Input their meal description and tap "Log Meal with AI". The macros will be calculated based on their health profile and saved under their record.

Viewing Family Dashboard Telemetry:

- Open the profile selector at the top-left of the main dashboard.
- Tap the avatar of your child or parent.
- The main dashboard, calorie rings, step tracker, and timelines will instantly transition to display that specific family member's data.

7. EMERGENCY CLINIC & HOSPITAL FINDER

Geolocation Emergency Querying:

- Go to Settings -> Nearby Clinics & Hospitals (or tap the Hospital icon in the Coach tab). Grant the app GPS location permission. The app queries the OpenStreetMap
- Overpass API to scan for medical centers.
- View the list of hospitals, clinics, and pharmacies sorted by proximity within a 5km to 15km radius.

Navigating to Facilities:

- Review the distance (in kilometers), phone contact link, and 24/7 emergency indicators on each hospital card.

- Tap "Get Directions" on your chosen facility. The app instantly opens Google Maps with pre-populated turn-by-turn routing to guide you safely.

8. GROCERY LIST & WEEKLY HEALTH REPORT

AI Smart Grocery List:

- Open Settings and tap "Smart Grocery List".
- The app aggregates your logged meals from the past 7 days, evaluates your macronutrient gaps, and compiles a categorized grocery list.
- The list recommends items in groups (Vitamins, Proteins, Grains) and flags items needed for your recommended healthy food swaps (e.g., listing whole-wheat flour to swap out white flour).

Exporting Weekly PDF Summaries:

- Open Settings and tap "Export Weekly Summary (PDF)".
- The app compiles your 7-day health metrics (average calorie intake, water totals, steps, active alerts, and AI dietitian comments) into a professional PDF document.
- Tap "Save PDF" to download it locally, print it, or share it with your doctor.

9. APP SETTINGS, LANGUAGE & ACCOUNT MANAGEMENT

Language Toggle:

- Open Settings and locate the Language section.
- Tap "Urdu" to instantly translate all menus, labels, and text fields into Urdu. Tap "English" to switch back.

Account Settings & Security:

- **Change Password:** Tap "Change Password", enter your current password, type your new password, and confirm to save.
- **Account Deletion:** Tap "Delete Account". Read the warning detailing that all records will be permanently removed. Enter your password and tap "Confirm Deletion" to purge all Supabase rows.

10. PREDICTIVE AI COACHING & NUTRITIONAL SWAPS

Accessing Predictive Coaching:

- Tap the "Coaching" tab in the bottom navigation bar.
- This screen displays your algorithmic 7-day habit score and healthy food swap recommendations.

Habit Score Analysis:

- View your large consistency ring showing your score out of 100. A score of 80+ indicates excellent compliance.
- Review the line chart illustrating your habit consistency over the past 7 days to track your progress.

Recommended Food Swaps:

- The screen highlights specific food swaps compiled by Gemini based on your logged meals (e.g., recommending brown rice instead of white rice).
- Each food swap card lists the suggested healthier ingredient alongside a clinical explanation detailing how the swap improves your blood sugar, cholesterol, or blood pressure.

11. AI-GENERATED CLINICAL WORKOUT PLANS

Viewing Daily Workouts:

- Tap the "Workout" tab in the bottom navigation bar.
- Tap "Generate Workout Plan" to have Gemini create a custom schedule based on your onboarding metrics and medical restrictions.
- Use the day-of-week tab bar selector at the top (Mon, Tue, Wed, etc.) to view the routine planned for each day.

Workout Walkthrough & Execution:

- View the overall focus of the day (e.g., "Strength", "Flexibility", "Rest").
- Tap the Warm-Up list to review warm-up routines.

- Swipe through the active exercise cards. Each card displays the exercise name, number of sets, target repetitions, rest interval, execution form tips, and clinical safety flags.
- As you complete each exercise, tap the circular checkbox to check it off.
- The progress bar at the top of the screen will update in real-time.
- Tap the Cool-Down list at the bottom to complete your cooldown stretch.

Regenerating Plans:

- If you switch to Ramadan Mode or update your medical profile, tap "Regenerate Workout" to recalculate your weekly training schedule to match your new state.